UNIT-II

MODERN ENGINEERING MATERIALS

1. SEMICONDUCTORS

Band theory of materials: The band theory of metals is based on the valence band and conduction band. It is also known as the band theory of solids. The gap or energy difference between the valence band and conduction band is called the forbidden gap. It defines conductors, semiconductors and insulators very clearly.

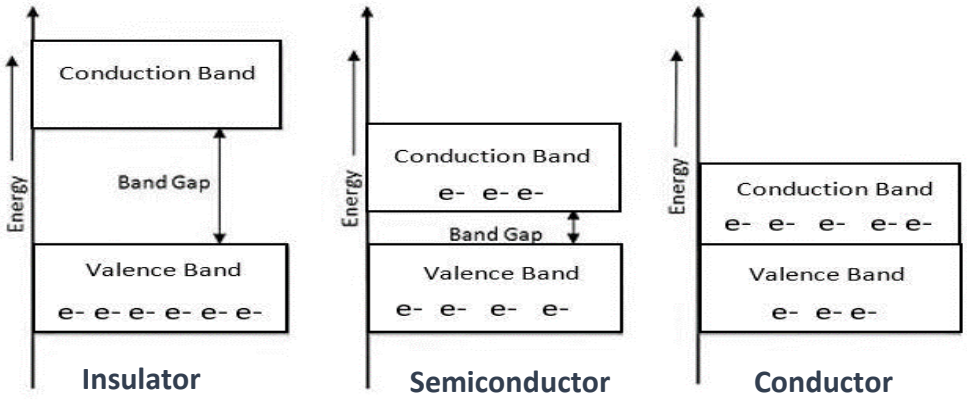
Conductors – Those materials which allow electricity to pass through them are called conductors. Examples – copper, iron, zinc etc.

Semiconductors – Those materials which show conductivity between conductors and insulators are called semiconductors. Examples – Si, Ge, As etc.

Insulators – Those materials which do not allow electricity to pass through them are called insulators. Examples – wood, glass, stones etc. After understanding all these terms now, you are ready to understand band theory.

In chemistry, according to the band theory of solids electrons jump from valence band to conduction band even at ordinary temperature and if this happens then the solid conducts electricity. Conductivity depends on the gap between the valence band and conduction band.

- If the gap or energy difference between the valence band and conduction band is more than or equal to 5ev then the material will behave as an insulator.
- If the energy difference between the valence band and conduction band is equal to or less than 3ev then the material or solid is called a semiconductor.
- If the valence band and conduction band overlap each other than the solid is called conductors.



The reasoning behind it is that electrons can jump from valence band to conduction band and thus conduct electricity. While if the gap is too much between the valence band and conduction band (more than or equal to 5ev) then electrons can't jump from the valence band to the conduction band, hence these materials or solids can't conduct electricity.

Semiconducting materials are classified into two types.

A) **Elemental semiconductors**: The elemental semiconductors are those compound of single species of atoms such as *Silicon(Si) Germanium(Ge)* and etc.

B) **Non elemental semiconductors:** If semiconductors are prepared by two or more different elements then those are called non elemental semiconductors. These are also called *inter-metallic or compound semiconductors*. These are formed by combination of *Group III & Group V or Group II & Group VI.* Non-elemental semiconductors are two types stoichiometry semiconductors and non-stoichiometry semiconductors.

Group-III	***Group-V***		***Group-II***	***Group-VI***
Al	P		Zn	O
Ga	As		Cd	S
In	Sb		Hg	Se
Ex: GaAS, GaP			**Ex: ZnO, CdS**	

Non-elemental semiconductors are commonly used in LED and electronic industry.

Semiconductors are also classifying as..

1. **Intrinsic Semiconductors:** Intrinsic Semiconductors are pure semiconductor it contains free from of (impurities) doping.It has low conductivity at room temperature.
   Ex: silicon and germanium
2. **Extrinsic Semiconductors:** Extrinsic Semiconductors are impure semiconductors it contains (impurities) doping. It hashigh conductivity at room temperature.
   Depending on the type of doping material used, extrinsic semiconductors can be two types.
   p-type semiconductors and n-type semiconductors.

Preparation of semiconductors: The semiconducting elements used for the application of semiconducting devices must be 99.999% pure i.e one of impurity for every 10^6 atoms of Si or Ge. To prepare such ultra-pure materials the following steps where involved.

1. Preparation of ultra-pure Si or Ge.
2. Preparation of single crystal of Si or Ge.

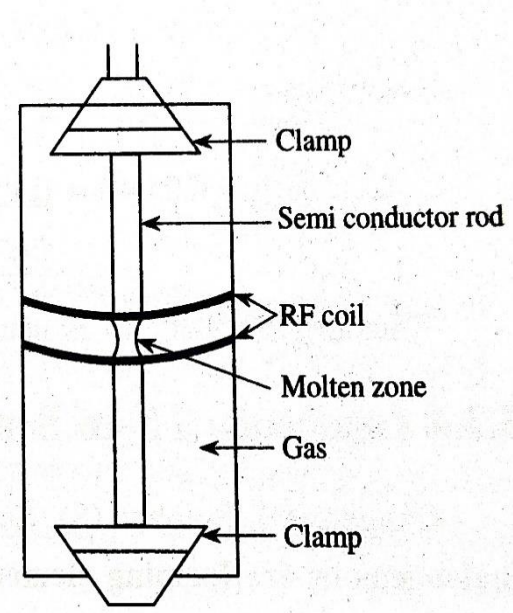
PREPARATION OF ULTRA PURE SILICA.

a) Distillation process: Metallurgical grade silica(MGS) is 98% pure. MGS is obtained by heating silica with coke. These silica contains impurities. For its purification silica is treated with HCl. Si will be converted to liquid trichloro silane ($SiHCl_3$). The $SiHCl_3$ is filtered and subjected to fractional distillation and pure vapors of $SiHCl_3$ are collected. And it is subjected to reduction reaction to get pure semiconductor grade silicon (SGS).

$$SiHCl_3 + H_2 \longrightarrow Si + 3HCl$$

Which is further purification, SGS is subjected to *Zone refining* process.

b) ZONE REFINING PROCESS:



Zone refining is a *metallurgical process* which is based on the principle that the impurities present in a metal are more soluble in molten metal then in solid metal. For the purification of silica *vertical zone refiner (show in fig.)* is used. A rod of silica to be purified is clamp and heated by *RF coil* heater to about $1000^{O}C$ in a reducing atmosphere. The heater moves from top to bottom very slowly. The total system filled with Argon gas to remove any impure gases. As the heater is move down the impurities also moves to bottom side of the rod. The process is repeated several times to reduce the impurity level and lower end of the road is removed after getting the desired purity because it is concentrated with impurities and finally only crystal in silica road is obtained to obtain the single crystal road we using process *Czochralski process*.

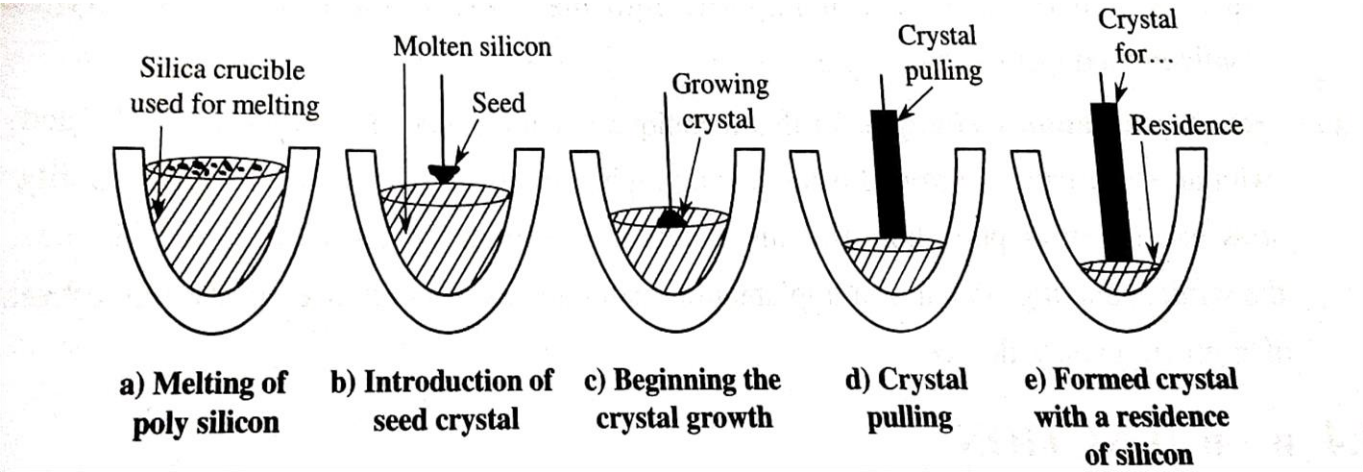
PREPARATION OF SINGLE CRYSTAL SILICA: The basic requirement for the fabrication of a semiconductor device that the semiconductor used must be a single crystal the preparation of silica and germanium procedures poly crystal in products that is crystals of various sizes. Hands single crystal of silica and germanium are produced by *Czochralski crystal pulling process*.

Czochralski crystal pulling technique:

In this method single crystals are grown in such a way that during crystal growth items reproduce the same crystal in arrangement as that of the seed crystal.

The process begins when the chamber which contains a crucible with silica is heated up to $1500^{O}C$. When silica is fully melt, a small seed crystal mounted on end of rotating shaft is slowly dips below the surface of the molten silicon. The shaft rotates in anti-clockwise and the crucible

rotates in clock wise direction. The rotating rod is than drawn upward very slowly about 25 mm per hour when making a crystal of silicon allowing in the form of boule to be formed. The boule can be from 1 to 2 meters depending on the amount of silica in the crucible. Careful an annealing process condition where adopted to remove oxygen. The above process is adopted even for the pure crystal pulling of Germanium.



Properties of semiconductors:

1) The resistivity of a semiconductor is less than an insulator but higher than a conductor.
2) Semiconductors show a negative temperature coefficient of resistance. In simple words, the resistance of the semiconductors decreases as the temperature increases and vice-versa.
3) At zero kelvin, semiconductors behave as insulators. As the temperature is increased it works as a conductor
4) The conductivity of the semiconductors increases when impurities are added. The process of adding impurities to semiconductors is called doping.

Applications of semiconductors:

1) Semiconductors are widely used in manufacturing electronics devices like transistors, diodes, sensors, integrated circuits.
2) Temperature sensors are made with semiconductor devices.
3) They are used in 3D printing machines.
4) Used in microchips and self-driving cars.
5) Used in calculators, solar plates, computers and other electronic devices.
6) Semiconductors are widely used in all electronic devices, like mobile phones, digital cameras, communication devices, trains, ATMs, etc.

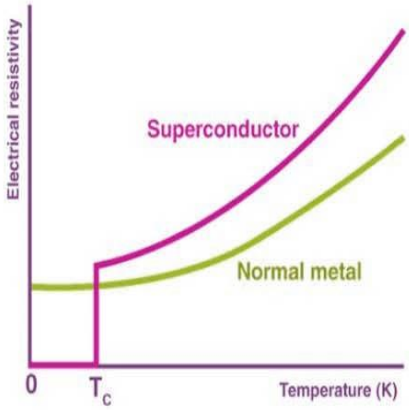
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2. SUPERCONDUCTORS

A solid which offers no resistance to the passage of electricity through it is called Superconductor. Superconductivity can be understood with help o quantum physics.

Ex: Lead, Tin, Hg, Nobium, magnesium diboride, cuprates such as yttrium barium copper oxide.

Superconductivity is exhibited by several metals, but only at low temperature. The temperature at with the normal metal passes into superconducting state is called the *superconducting transition state or critical temperature*, denoted by **Tc**. The resistivity – temperature behaviors for superconductive and non-superconductive materials are differentiated show in the fig.



Types of Superconductors: Superconductors are divided into two types based on their *magnetic* response.

Type-I or Ideal or soft Superconductors & Type-II or Hard Superconductors.

Type-I Superconductors (Ideal Superconductors/ Soft superconductors):

a) Type-I superconductors are also called as ideal Superconductors.

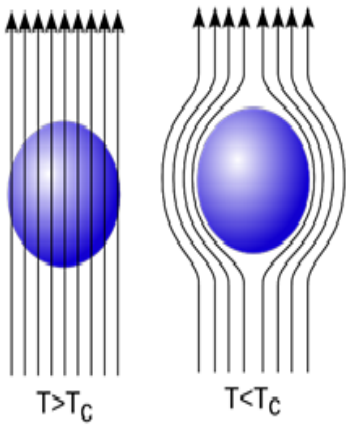
b) When the Type I superconductor is placed in the magnetic field, it suddenly or easily loses its superconductivity at the critical magnetic field (H_c)

c) In the superconducting state, these materials become diamagnetic.

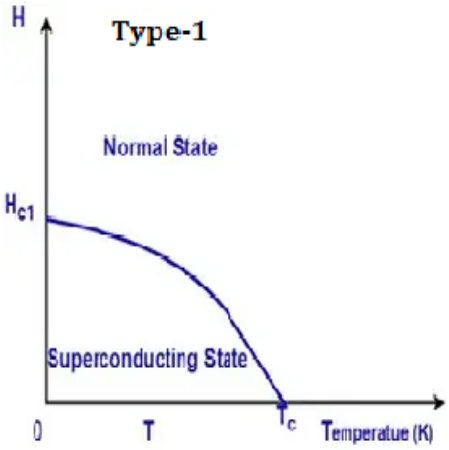
d) They exhibit the "Meissner effect", which is the expulsion of magnetic flux from their interior as they transition to the superconducting phase.

e) These are also known as "*soft superconductors"* because of this reason that is they lose their superconductivity easily.

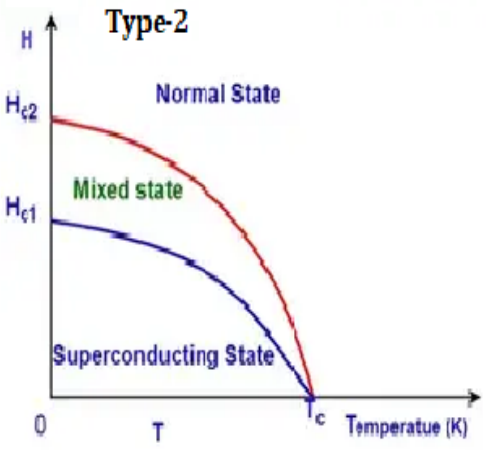
Ex: Lead (Pb), Mercury (Hg), Tin (Sn), etc.



Meissner effect



Variation in Critical magnetic feild with Temparature for TYPE-I Superconductor



Variation in Critical magnetic feild with Temparature for TYPE-II Superconductor

Type-II Superconductors (**Hard superconductors**):

a) Type-II superconductors lose their superconductivity gradually but not easily when placed in the external magnetic field.
b) Type-II superconductors start to lose their superconductivity at the lower critical magnetic field (H_{C1}) and completely lose their superconductivity at the upper critical magnetic field (H_{C2}).
c) The state between the lower critical magnetic field (H_{C1}) and upper critical magnetic field (H_{C2}) is known as *vortex state or intermediate state*.
d) They are also known as "hard superconductors" because they lose their superconductivity gradually but not easily. Type II superconductors obey the Meissner effect but not completely. Ex: Nb-Sn, Nb-Ti, etc.

Preparation of Superconductors:

Yttrium barium cuprate (YBa_2,Cu_3O_{7-x}, where x is small)

This functions as a superconductor at a critical temperature (T_C) of 90 °K. The constituents of this material i.e., yttrium, barium and copper are in 1:2:3 molar stoichiometric ratio and hence are called as *1:2:3 superconductors.*

Synthesis of 1: 2: 3 superconductor by ceramic method: $YBa_2\ Cu_3\ O_{7-x}$ can be synthesised by solid state ceramic technique. This involves the following steps:

a. Preparation of a homogeneous mixture of there oxides *(Y_2O_3, $BaCO_3$ and CuO)* in their molar ratios.

b. Heating them to obtain oxygen deficient superconductor in a muffle furnace.

c. Annealing the above compound to room temperature to retain its composition, structure and superconducting properties.

Properties of Superconductors:

1. Superconducting materials are generally brittle in nature.
2. At room temperature, the resistivity of superconducting material is greater than other elements.
3. A Superconductor exhibits perfect diamagnetism.
4. Thermoelectric effects of materials disappear in the superconducting state.
5. If strong magnetic field is applied below T_C. Its superconducting property is destroyed.

Applications of Superconductors:

1. Superconductors can be used to perform logic and storage functions in computers.
2. Superconducting magnets capable used for magnetic resonance imaging (MRI) in the medical field as a diagnostic tool.
3. Some superconductor's excellent catalysts for industrial chemical processes.
   Ex: $YBa_2Cu_3O_7$act as catalysts in oxidation dehydrogenation reactions
4. Superconducting material used in manufacture of electrical generators and transformers
5. Superconducting magnets are used for operating frictionless, high-speed trains.

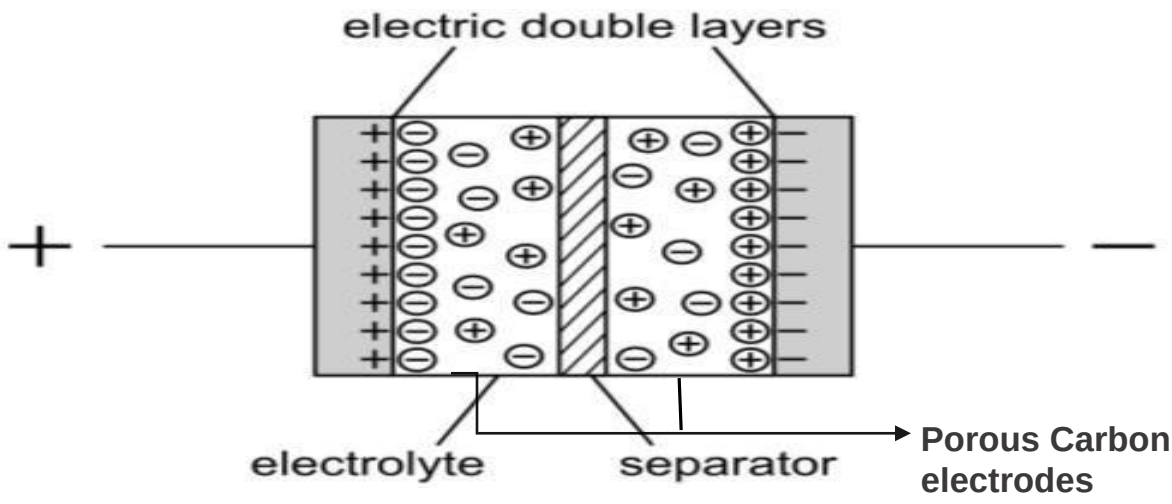
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3. SUPERCAPACITORS

Supercapacitor also called as *Ultracapacitor*. That can store large amounts of energy nearly 10 to 100 times more energy when compared to the normal capacitors. It is widely preferred than batteries because of its faster charging capacity and faster delivery of energy. It has more charging and discharging cycles than rechargeable batteries. The main advantage of this capacitor is its efficiency and high-energy storage capacity.

Construction of Supercapaciter:

- Supercapacitors are constructed by two electrodes that are made up of porous active carbon coating or carbon rods. The electrodes are coated with aluminum which serve as current collectors. The electrodes with current collectors are immersed in an electrolyte.
- When Voltage is applied, the positive ions from the electrolyte moves towards the negative electrode and the negative ions from the electrolyte moves towards the positive electrode.
- The forces of attraction between the electrode and electrolyte is electrostatic forces and an electrical double layer formed at the two electrodes. And develops the charges. The separator prevents the charge in moving between the electrodes.



Types of Supercapacitors.

Supercapacitors are classified into three types:

- **Electrostatic Double Layer Capacitors(EDLC):** Two electrodes, a separator, and an electrolyte are components of these kinds of capacitors. The mixture of positive and negative ions dissolved in solution is known as an electrolyte. A separator separates the two electrodes. The electrostatic double-layer capacitance of the carbon electrodes used in these supercapacitors is significantly higher. Electrostatic double-layer capacitors have a smaller charge separation than normal capacitors; it is between 0.3 and 0.8 nm.

- **Pseudo Capacitors:** Electrochemical pseudo-capacitors are another name for pseudo-capacitors. These capacitors use conducting polymer electrodes or metal oxide electrodes with substantial electrochemical pseudo capacitance. By transferring electron charges between an electrode and an electrolyte, these components store electrical energy. A reduction-oxidation reaction, or redox reaction, can accomplish this.
- **Hybrid Capacitors:** The double-layer capacitor and pseudo-capacitor techniques are used to create the hybrid capacitors. Different electrodes with various properties are utilized in these components. The ability of one electrode to display electrostatic capacitance and the ability of the other electrode to display electrochemical capacitance. A few instances of hybrid capacitors are lithium-ion battery capacitors.

Advantages:

* Provide Backup power and peak power.
* Enhance run time of battery and prolong its life.
* Helps to reduce size, weight and cost of battery.
* It has unlimited life cycles (charge & discharge)
* Environmental friendly.
* Less space requirement achieved.

Disadvantages:

* They discharge themselves more frequently.
* Low voltages exist within individual cells.
* It cannot be used in AC and higher frequency circuits.
* They have higher self-discharge rate. This is considerably high compare to battery.

Application of Supercapacitors:

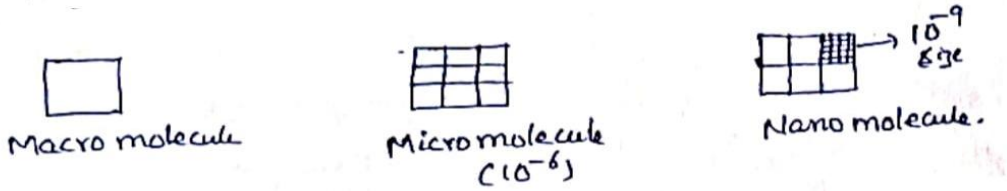
Due to unique storage capability of supercapacitors, it has been widely used in various applications like Electric drives, UPS, Electric vehicles, SSD's, LED flashlights etc. Let us discuss few of them. Supercapacitors are used in the following:

1.Electric cars
2. Wind turbines
3. Photographic flash
4. Flywheel in machines ayers
5. Static memories (SRAM).
6. Industrial electrical motors.
7. Regenerative braking in the automotive industry.

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4. NANOMATERIALS

Nano means Dwarf (too small). It means a billionth part of any unit. A Nano second is one billionth of a second. And nanometer is one billionth part of a meter. i.e. 1mm = 1/1000000000 or 10^{-9} of a meter.

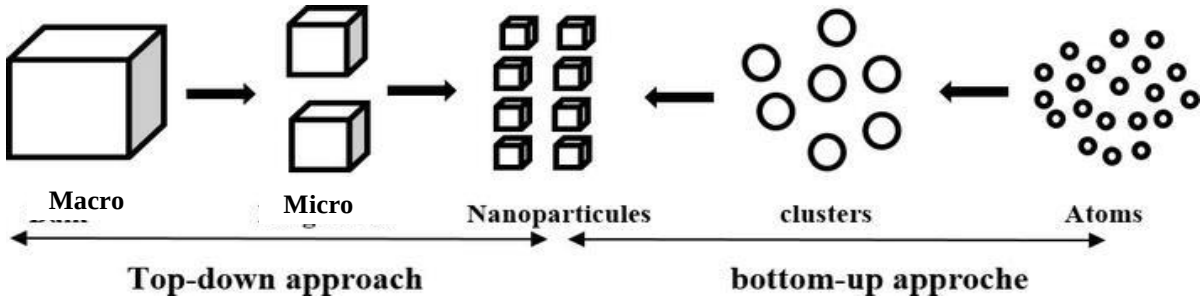


Macro molecule can be divided to form micro molecule. i.e. 10^{-6}mt. This micro molecule divided no. of times to form a Nano molecule. i.e. 10^{-9} meter. Those Nano molecules we can't see directly. They have to observed/study with different microscopes like

** Atomic force microscope (AFM)* ** Magnetic force microscope (MFM).*

Basically two major types of approach for the preparation of Nanomaterials.

a) Top-down approach & b) Bottom-up approach.



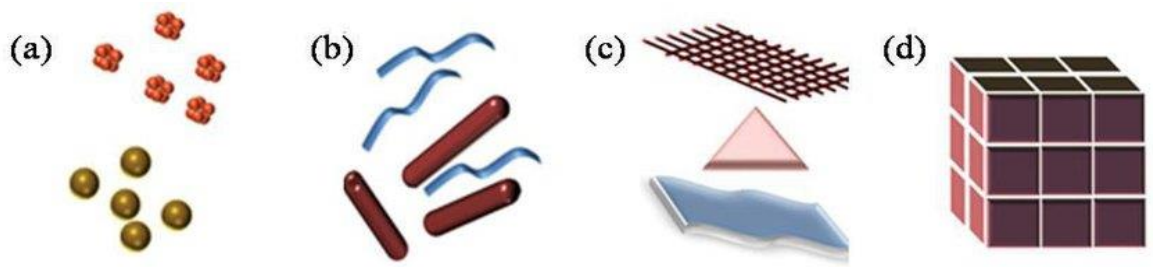
Calcification of Nanomaterials: Depending upon size nonmaterial classified 4 types.

a) **Zero dimensional:** Nano materials that have all three dimensional x, y and z with in the Nano scale below 10nm. *Ex: Quantum dots, Nano dots and Fullerenes.*

b) **One dimensional:** Nano materials that have any two dimensional x, y and z with in the Nano scale above 10nm. *Ex: Nano tubes, Nano rods and Nanowires*.

c) **Two dimensional:** Nano materials that have two dimensional our said with in the Nano scale 1nm to 100nm.*Ex: Graphene and Nano layers.*

d) **Three dimensional:** Nano materials that cannot be confined to the nano scale in any dimensional. *Ex: Graphite and Diamond.*



Properties of Nanomaterials:

1. It has high thermal conductivity
2. It has high electrical conductivity
3. It has high melting point.
4. It has highly flexibility
5. It has very strong and light weight
6. It has transparent material
7. It has low density and high strength.
8. It has more chemical reactivity.

Application of Nanomaterials:

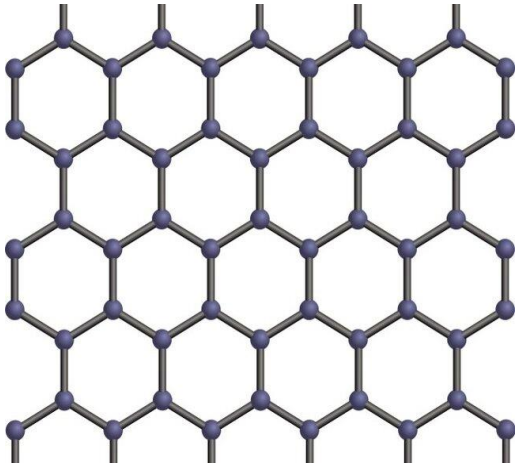
1. It used as solar cells
2. It used as air and water filtration
3. It used as Prevent HIV virus.
4. It used as cancer treatment.
5. It used as Drug delivery system in Medical
6. It used as aerospace applications
7. It used as conductors.
8. It used as bio, gas, temperature and environment sensors.

GRAPHENE:

Graphene is allotrope of carbon discovered in 2003.It is a honeycomb like structure. Graphene is single planer sheet of carbon atoms. All carbon atoms arranged hexagons. Each carbon atom sp^2 hybridized and each sp^2 carbon is attached to 3 other carbon atoms.

Graphene has a very good conductivity property Because of the un hybridized p-orbital electron. Graphene structure with 0.335nm.3 million graphene sheets placed in 1 millimeter thickness. This sheet can be folded to give carbon nanotubes.

Preparation: The Arc-discharge method is used to synthesis Graphene. High current is applied between two graphite (carbon) electrode in an inert atmosphere.



Graphene sheet

Properties of Graphene

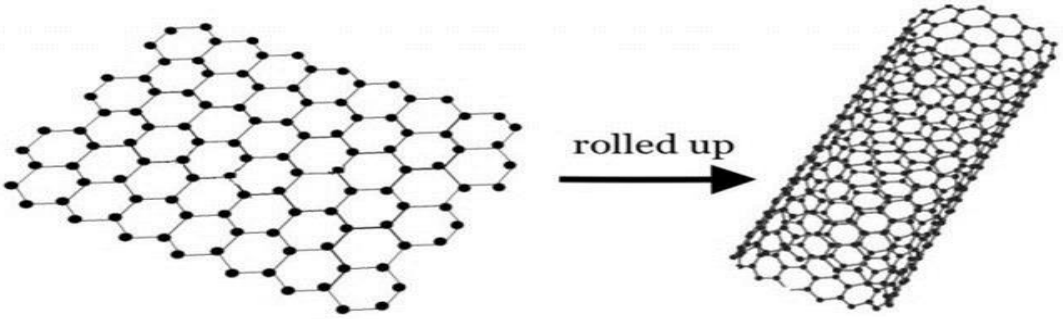
1. High thermal conductivity
2. High electrical conductivity
3. High elasticity and flexibility
4. High toughness
5. High resistance
6. Transparent material
7. Expose to sun light to generate electricity.

Application of Graphene

1. Graphene used as solar cell
2. Graphene used as batteries
3. Graphene used as nuclear power plants
4. Graphene used as drug delivery
5. Used as biomedical applications.
6. Graphene used as Prevent HIV virus.
7. Graphene used as bio sensors.

CARBON NANOTUBES (CNT):

Carbon nanotubes are discovered by sumio Ijima in 1991. A single sheet of graphite is called graphene. A graphene sheet which has been rolled up to cylindrical shape is called carbon nano tubes.



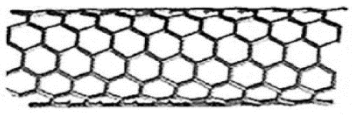
Graphene sheet Carbon Nanotube

carbon nanotube classified into *two* types based on number of layers.

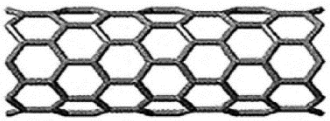
1) **Single-walled nanotubes (SWNT):** A single walled nanotube made by only one graphene sheet is use to form the nanotubes.

SWNT are also classify in to three types based on its structural properties.

a) Armchair b) Chiral c) Zigzag

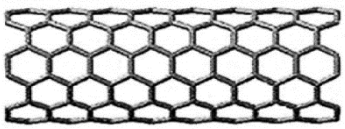


Chiral



Zigzag

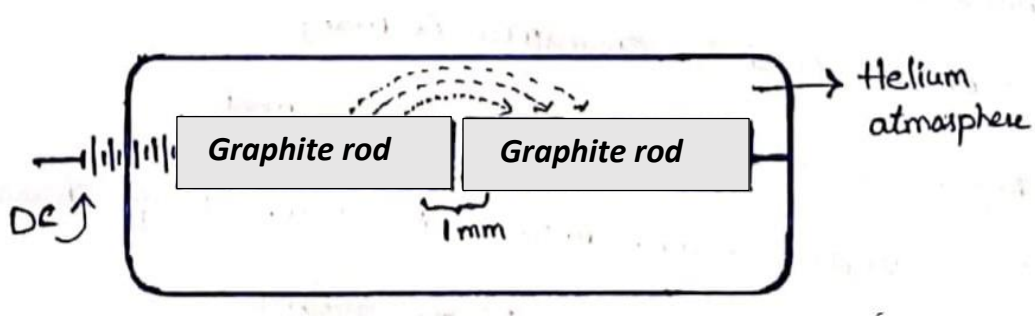
2) **Multi-walled nanotubes (MWNT):** A multi walled carbon nanotube made by more than one graphene sheet is use to form the nanotube.



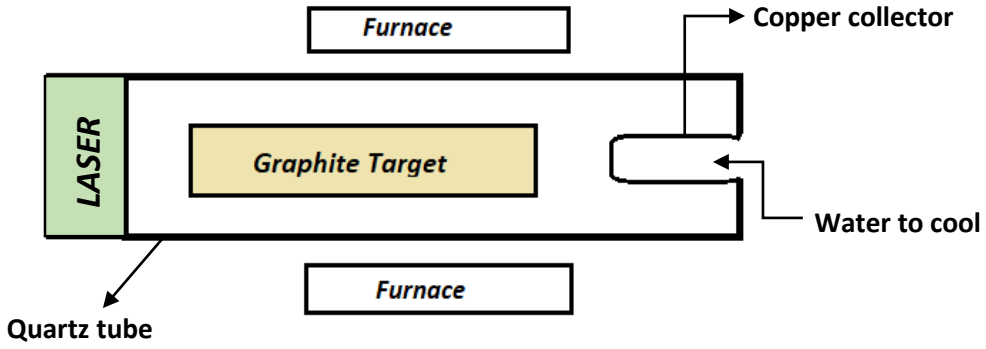
Armchair

Preparation of Carbon Nanotubes:

1. Arc-discharge method: In this method, both single-walled and multi-walled carbon nanotubes are prepared. Two graphite rods are placed in a helium atmosphere with a 1mm distance between them. A direct current passes through the graphite rods, creating high temperatures. One of the carbon rod discharges like soot (carbon atoms) and deposits on the surface of the other carbon rod, creating tiny rod-shaped CNTs.



Laser Ablation method: In laser ablation process, the graphite target is vaporized by a laser source. In this method, the graphite target is placed at the center of quartz chamber filled with argon gas and maintained at 1200°C. The graphite target is vaporized by continuous laser source. The vaporized target atoms (carbon) are swept toward a cooled copper collector by the flow of argon gas. The carbon atoms are deposited and grown as carbon nanotubes (CNTs) on the cooled copper collector. This process allows for the production of single-walled and multi-walled carbon nanotubes. In this process, catalyst nanoparticles of Fe, Co, Ni are used.



Properties of Carbon Nanotubes:

1. Carbon nanotubes are stiff. They are as stiff as a diamond (the hardest natural material in nature).
2. The density of the carbon nanotubes is one-fourth of that of steel.
3. Carbon nanotubes are stronger than steel. These are ten times stronger than steel.
4. They exhibit extraordinary mechanical properties and CNTs are flexible, elastic.
5. Carbon nanotubes have a high thermal capacity and good conductors of heat.
6. Carbon nanotubes have good electrical conductivity.
7. CNTs are chemically neutral. So, they are chemically stable. Therefore, carbon nanotubes resist corrosion.

Applications of CNTs:

1. CNT's are used as the strengtheners of Composite materials.
2. CNT's are used as tips for analysis of DNA. CNTs serve as catalyst supports.
3. CNT's are used as heat conductors, and solar cells,
4. CNT's compositions used for aerospace applications
5. CNT's are used as store hydrogen to be used as fuel
6. CNT's are used in Drug delivery system in Medical.
7. CNT's are used for Air and water filtration.
8. CNT's are used in textile industry such as antibacterial Coatings.
9. CNT's are used for making paper batteries. CNTs can be used in anti-corrosion coatings.
10. CNTs are used in supercapacitors and batteries to store and release energy efficiently.

FULLERENES (C_{60}):

Fullerene is allotrope of carbon. Ii is discovered by H.Kroto and R.E Smelley in 1935. Fullerene structure developed Buck minister fuller, so it is called buck minister fullerene.Fullerene has hollow spherical Soccer ball like structure. So it is called Bucky ball. Fullerene in represented by C_{60}.



C_{60} contains 60 carbon atoms these are arranged by 32 rings. i.e., *12 pentagons* (5 member rings) and *20 hexagons* (6 member rings). Each pentagon covered with 5 six hexagons. And each carbon of fullerene is sp^2 hybridization and C-C band dough is $1.4A^0$.

Preparation of Fullerenes:

The A*rc discharge* method is used to synthesis fullerenes. High current is applied between two graphite (carbon) electrode in an inert atmosphere (helium or argon). In this method, the carbon electrode gets evaporated. The soot is generated and collected on cooled surfaces. The obtained soot is a mixture of fullerenes. These mixed fullerenes are separated by solvent extraction process and chromatography techniques.

Properties Fullerene:

1. Fullerene is black powder material.
2. Fullerene is ferromagnetic.
3. Fullerenes is high conductivity.
4. Fullerene is tough and thermally stable.
5. Fullerene is soluble in solvents like Toluene, Benzene and insoluble water
6. Fullerenes when doped or crystallized with alkali or alkaline earth metals showcases superconductivity properties.

Application of Fullerene:

1. Fullerene used as biological Antioxidants.
2. Fullerene used as gas and temperature sensor's
3. Fullerene used as Prevent HIV virus.
4. Fullerene used as solar cells and conductors.
5. Fullerene used as lubricant.
6. Fullerene used as absorbent for gas.

******ALL THE BEST******